

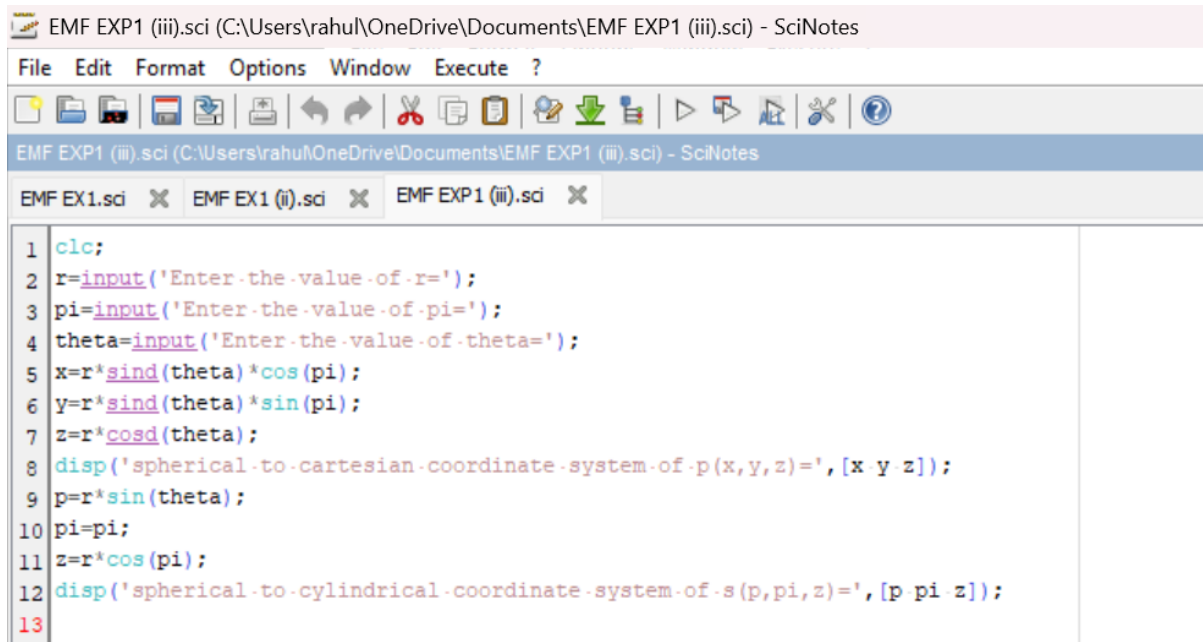
RAHUL R – 192412255

ECA0502 EMF

EXP 2 TESTCASE 2

Transformation between Cartesian, Cylindrical and spherical coordinate system

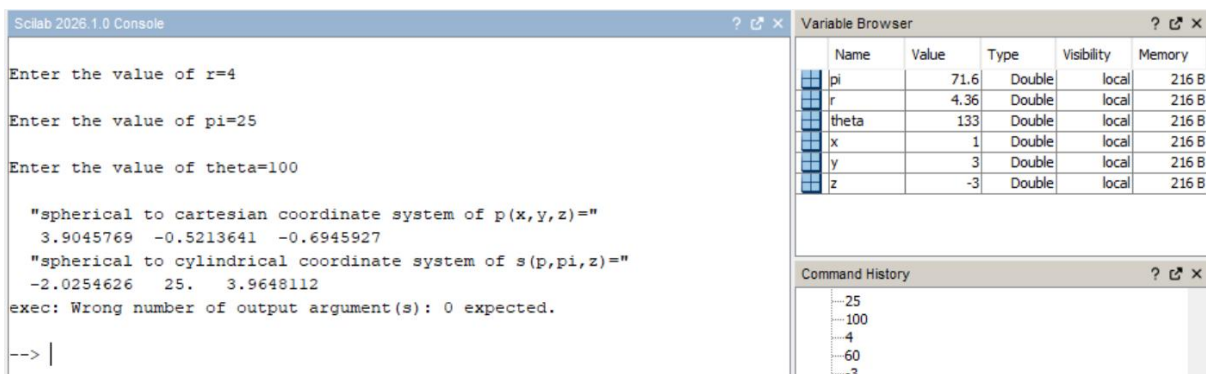
PROGRAM:



The screenshot shows a SciNotes window titled "EMF EXP1 (iii).sci (C:\Users\rahul\OneDrive\Documents\EMF EXP1 (iii).sci) - SciNotes". The menu bar includes File, Edit, Format, Options, Window, Execute, and ?. The toolbar contains various icons for file operations and execution. The script content is as follows:

```
1 clc;
2 r=input('Enter the value of r=');
3 pi=input('Enter the value of pi=');
4 theta=input('Enter the value of theta=');
5 x=r*sind(theta)*cos(pi);
6 y=r*sind(theta)*sin(pi);
7 z=r*cosd(theta);
8 disp('spherical to cartesian coordinate system of p(x,y,z)=[x y z]');
9 p=r*sin(theta);
10 pi=pi;
11 z=r*cos(pi);
12 disp('spherical to cylindrical coordinate system of s(p,pi,z)=[p pi z]');
13
```

OUTPUT:



The screenshot shows the SciLab 2026.1.0 Console and Variable Browser. The console displays the input values and the resulting coordinate transformations.

Console Output:

```
Enter the value of r=4
Enter the value of pi=25
Enter the value of theta=100

"spherical to cartesian coordinate system of p(x,y,z)="
3.9045769 -0.5213641 -0.6945927
"spherical to cylindrical coordinate system of s(p,pi,z)="
-2.0254626 25. 3.9648112
exec: Wrong number of output argument(s): 0 expected.
--> |
```

Variable Browser:

Name	Value	Type	Visibility	Memory
pi	71.6	Double	local	216 B
r	4.36	Double	local	216 B
theta	133	Double	local	216 B
x	1	Double	local	216 B
y	3	Double	local	216 B
z	-3	Double	local	216 B

Command History:

```
--25
--100
--4
--60
--3
```

OBSERVATION:

test case (iii) $r = 4$
 $\phi = 25^\circ$
 $\theta = 100^\circ$

spherical $\rightarrow$ cartesian

$$x = r \sin(\theta) \cos \phi$$

$$\sin(100) = 0.9848078$$

$$\cos(25) = 0.992028$$

$$x = 4(0.9848078)(0.992028)$$

$$\boxed{x = 3.9042}$$

$$y = r \sin(\theta) \sin \phi$$

$$y = 4(0.9848078)(0.1323518)$$

$$\boxed{y = -0.5214}$$

$$z = r \cos \theta$$

$$= 4(-0.173648)$$

$$\boxed{z = -0.6946}$$

spherical $\rightarrow$ cyl

$$\rho = r \sin \theta$$

$$= 4 \sin 100 = -2.0255$$

$$\boxed{\rho = -2.0255}$$

$$\boxed{\phi = 25^\circ}$$

$$z = r \cos \phi$$

$$= 4(0.992028)$$

$$\boxed{z = 3.9648}$$

$$(x, y, z) = (3.9042, -0.5214, -0.6946)$$

$$(\rho, \phi, z) = (-2.0255, 25^\circ, 3.9648)$$